MTH404 - Dynamics (Quiz 4)

Question # 10 of 10 (Start time: 04:19:39 PM, 07 July 2025)

The central force is ---------- if it is directed towards the centre.

Select the correct option

○	none of these
◉	attractive
○	not attractive
○	repulsive

MTH404 - Dynamics (Quiz 4)

Question # 4 of 10 (Start time: 04:16:50 PM, 07 July 2025)

The analytical condition for a point with radius vector r to be an apse is dr/dtheta=

Select the correct option

○	-1
●	0
○	1
○	-2

MTH404 - Dynamics (Quiz 4)

Question # 9 of 10 (Start time: 04:19:14 PM, 07 July 2025)

The central force is ---------- if it is directed away from the centre.

Select the correct option

○	not attractive
◉	repulsive
○	attractive
○	none of these

MTH404 - Dynamics (Quiz 4)

Question # 8 of 10 (Start time: 04:18:39 PM, 07 July 2025)

The maximum horizontal range of a projectile is 80ft, the velocity of the projection is-----.

Select the correct option

◉	50.6 ft/s
○	40.2 ft/s
○	35 ft/s
○	45.5 ft/s

MTH404 - Dynamics (Quiz 4)

Question # 2 of 10 (Start time: 04:16:07 PM, 07 July 2025)

In a conservative field, the total energy of a particle through the motion

Select the correct option

○	none of these
○	varies rapidly
●	remains constant
○	does not remain constant

MTH404 - Dynamics (Quiz 4)

Question # 5 of 10 (Start time: 04:17:07 PM, 07 July 2025)

iN defining orbital velocity, thing whis is unimportant is --------.

Select the correct option

○	gravitational force
○	mass of Earth
○	radius of Earth
●	mass of satellite

MTH404 - Dynamics (Quiz 4)

Question # 3 of 10 (Start time: 04:16:21 PM, 07 July 2025)

The maximum horizontal range of a projectile is 100ft, the velocity of the projection is-----.

Select the correct option

○	5.6ft/sec
○	4.6ft/sec
○	3.6ft/sec
◉	56.57ft/sec

MTH404 - Dynamics (Quiz 4)

Question # 1 of 10 (Start time: 04:15:20 PM, 07 July 2025)

If $45^0$and 20 are the angle and velocity of projection respectively, then the time of flight of the projectile is .....

Select the correct option

○	$20\sqrt{2}g$
○	$20\sqrt{2}$
○	20g
◉	$\frac{20\sqrt{2}}{g}$

MTH404 - Dynamics (Quiz 4)

Question # 7 of 10 (Start time: 04:17:33 PM, 07 July 2025)

If 45^0 and v_0 are the angle and velocity of projection respectively, then the time taken to attain the maximum height of the projectile is .....

Select the correct option

○	$v_0\sqrt{2}g$
◉	$\frac{v_0}{\sqrt{2}g}$
○	$\frac{v_0}{\sqrt{2}}$
○	$v_0\sqrt{2}$

Click to

Question # 6 of 10 (Start time: 04:17:21 PM, 07 July 2025)

The orbit described under a central attractive force varying directly as the distance is --------- having the centre at the centre of the force.

Select the correct option

- ○ none of these
- ○ a circle
- ○ a straight line
- ◉ an ellipse

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